

TrES-1b

R-2416

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_110831 · created 2026-08-21T23:12:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455804.78702 +/- 0.00079 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.087 +/- 0.052
Transit depth (Rp/Rs)^2	0.76 +/- 0.91 [%]
Orbital Inclination (inc)	85.03 +/- 1.46
Airmass coefficient 1 (a1)	1.02 +/- 0.14
Airmass coefficient 2 (a2)	-0.032 +/- 0.067
Scatter in the residuals of the lightcurve fit is	13.62 %
Best Comparison Star	#1 - [529, 212]
Optimal Aperture	3.51
Optimal Annulus	8.18
Transit Duration (day)	0.045 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.087 ± 0.052 vs 0.1358 (deviation 0.84σ)
Duration fit vs published	0.045 d vs 0.1042 d (deviation 1.61σ)
Mid-time O–C	-1.2 min
Depth SNR	1.5
Residual scatter	13.62 %

Light curve

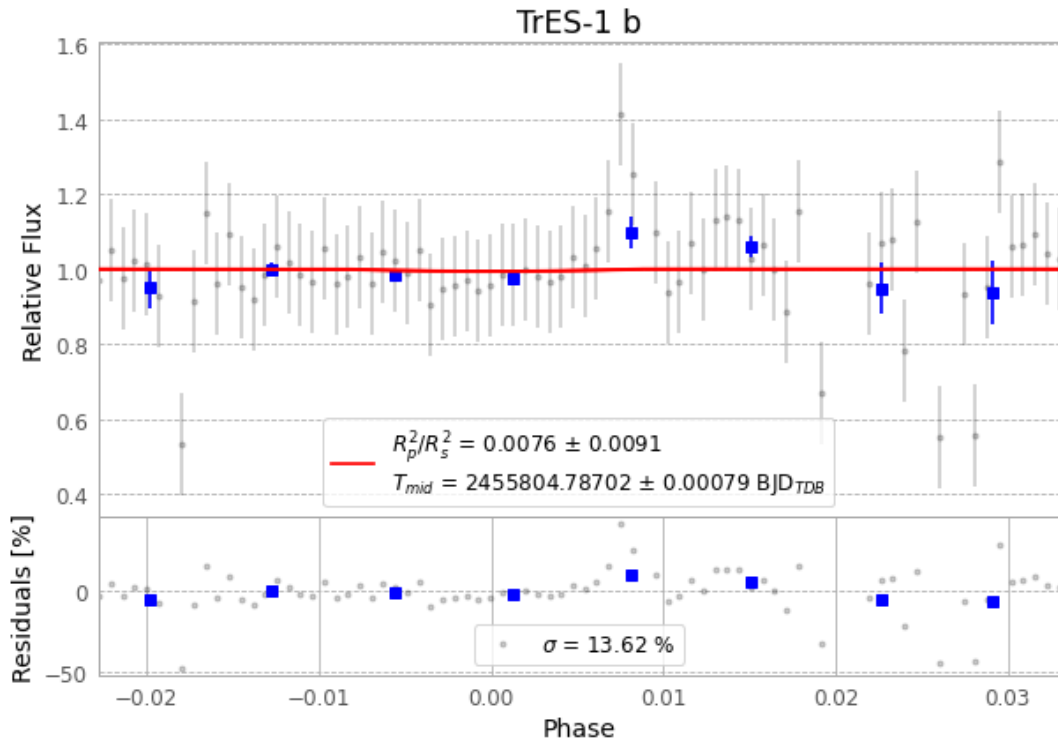


Figure 1 — FinalLightCurve_TrES-1 b_2011-08-30.png (SHA-256 6b763daeac892a30...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [240, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cd82d001ecc13581...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

82 FITS frames (54 MB), TRES-1110831050912.FITS → TRES-1110831091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-110831.log (c17402d2a8de...), FinalLightCurve_TrES-1 b_2011-08-30.png (6b763daeac89...), FinalLightCurve_TrES-1 b_2011-08-30.csv (668a58aee2f6...)

Compute resources

Wall time 154.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:12Z.